

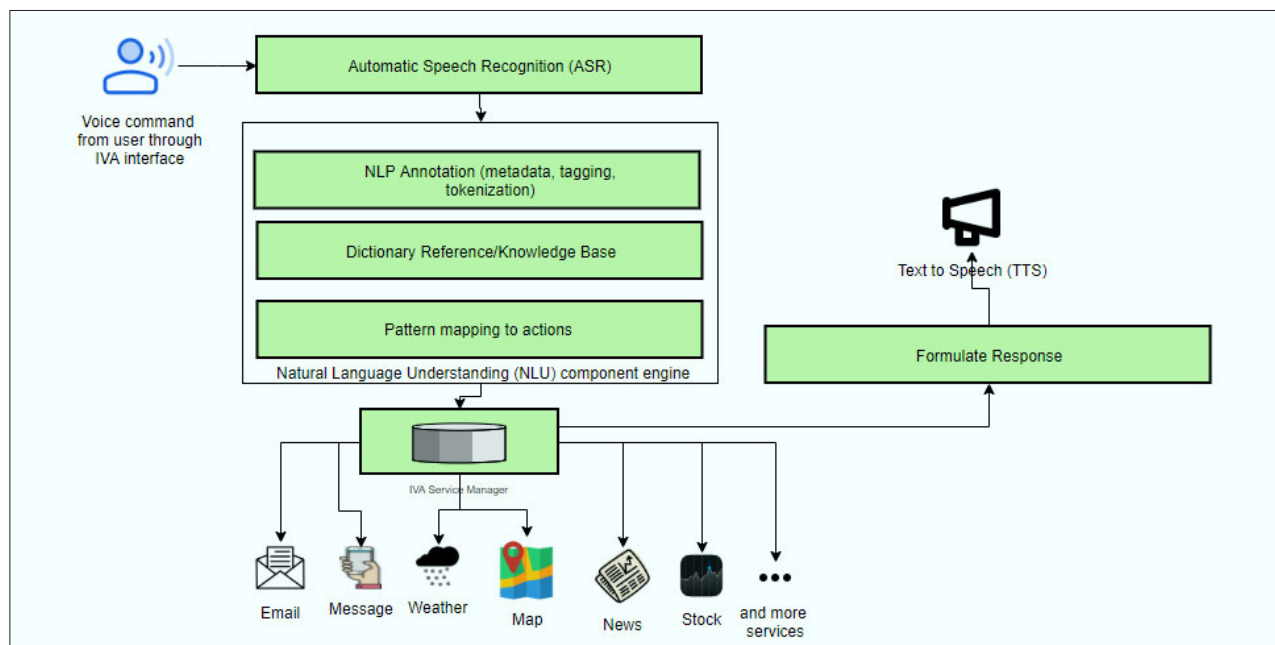


Figure 1: Execution steps of IVA

Chatbots vs conversational AI

Telemedicine and virtual health consultation are the new normal in the world after the recent pandemic. Hence, small clinics to large medical institutions prefer to develop and deploy a health bot, which can help patients with remote consultation. Health bots typically use AI and ML to process the query written by users through NLP, search for the response from their knowledge base, and have an interactive discussion with them. Hence, they are termed conversational AI enabled health bot solutions.

Chatbots and contact center AI play a critical role in developing a better customer experience. Traditionally, conversational AI was built by training the system to build the knowledge base and worked with a concrete set of functionalities only. With modern AI/ML services, self-managed conversational AI applications can be built very easily.

Azure Bot Service offers an AI agent that interacts with humans for support activities such as virtual banking assistance, insurance advice, IT helpdesk support and medical consultation, to

name a few. The conversational AI solutions it offers are built on the following technical components.

Bot logic, which includes language understanding (LUIS) for predicting user intent from conversational natural language, QnA Maker to prepare questions and answers for FAQs and quick help, and Conversation Learner to automatically build the knowledge base, based on example interactions.

Speech capture converts speech to text using specific vocabulary and by understanding various styles of speech. Machine translation is used to translate text from different languages supported by the bot service.

Speech synthesis prepares the response from the knowledge base and pre-built answers, and uses a text to speech service to speak out responses from pre-built voices or create a custom voice. It helps users feel they are interacting with human assistants.

Some common use cases in conversational AI

Conversational AI is one of the most promising sectors of AI, with the global market for it projected to grow

from US\$ 4.2 billion in 2019 to US\$ 15.7 billion by 2024. It is being used in both the commercial and domestic sectors today.

Here are a few use cases that highlight the revolution that conversational AI truly is.

Customer service: Conversational AI delivers the most impactful customer support services. It is now not only concerned with helping customers find timely solutions but is also making the life of agents better. Here are some examples.

- **Better agent experience:** AI helps agents do their jobs better; with the help of conversational AI, support executives can get through tickets much faster and have high quality interactions with customers.
- **More self-service:** By using a tool that can better leverage the knowledge base and company data, organisations can help customers help themselves better than ever before. AI can help answer the most frequently asked questions, giving customers what they need and deflecting redundant tasks from support agents.